

**WEBB INDUSTRIES' DC MOTOR
ACHIEVES SPEC-COMPLIANT
POSITION CONTROL WITHIN RANGE
-4.81 TO +5.81 V (DEAD-ZONE
-0.078 TO +0.0488 V), FOR
SETPOINT STEPS ≥ 1.0 RAD, AND
PID CONTROL IS RECOMMENDED**

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Introduction

Webb Industries plans to use a new direct-current motor as a positioning actuator and engaged Burdell Inc. to verify its dynamics and deliver a controller within given specifications. Results show a motor gain of $K_m \approx 55.9967 \pm 0.061 \text{ rad}\cdot\text{s}^{-1}\cdot\text{V}^{-1}$ ($K=2$) and a time constant of $T_m \approx 0.082 \pm 0.003 \text{ s}$ ($K=2$) in the unloaded configuration, increasing to $T_m \approx 0.216 \pm 0.012 \text{ s}$ ($K=2$) with a +104g load. Reliable operation is bounded by a dead zone of -0.078 to $+0.0488 \text{ V}$ and saturation beginning at -4.81 V and 5.81 V , yielding a recommended linear command window of approximately ± 0.05 to $\pm 5 \text{ V}$; in addition, responses to small setpoint changes below 1.0 rad are unreliable due to dead-zone and quantization effects. Using these parameters, proportional (P), proportional-derivative (PD), and proportional-integral-derivative (PID) controllers were designed and evaluated against $M_p \leq 6\%$, $t_s \leq 0.5 \text{ s}$, and $|e_{ss}| \leq 0.25 \text{ rad}$. All three controllers meet the specifications within the linear window; PID provides the fastest compliant response, while PD offers a low-overshoot alternative with slightly longer settling. These findings support commissioning the motor for position control, with PID recommended as the default and PD preferred when overshoot must be minimized.

Methodology

To develop a controller that meets the required performance specifications, a multi-stage methodology was employed to first characterize the motor, then use that empirical model to develop and validate a high-performance controller. The initial phase focused on experimentally determining the motor gain constant (K_m) and time constant (T_m) for the first-order transfer function model proposed by Webb Industries (Equation 1). This was accomplished using both time-domain and frequency-domain analyses. The resulting parameters were then used to calculate initial PID gains, which were first tuned in a non-linear Simulink model incorporating the motor's measured characteristics. The final stage involved implementing the tuned controller on the physical hardware to validate the model's predictive accuracy and verify that performance specifications were met.

$$\frac{\Omega(s)}{V_{in}(s)} = \frac{K_m}{(T_m s + 1)} \quad [1]$$

The time-domain analysis was conducted by measuring the motor's transient response to a series of step voltage inputs. Using LabView, constant DC voltages were applied, and time-series data for both the input voltage and the resulting angular velocity were recorded. The motor gain constant (K_m) was calculated as the slope of a York fit to the steady-state velocity (ω_{ss}) versus input voltage (V_{in}) data. The motor time constant (T_m) was determined by averaging the time required to reach 63.2% of the final steady-state velocity for all tests within the linear range. To investigate the effect of increased inertia on system parameters, this entire procedure was performed with and without an additional 104g mass.

Frequency-domain analysis was performed by driving the motor with a sinusoidal voltage input at discrete frequencies ranging from 0.1 Hz to 10 Hz. For each frequency, the steady-state portion of the time-series data was isolated, and a sinusoidal model was mathematically fitted to determine the amplitude and phase of both the input and output signals. This data was used to generate a Bode plot, from which the motor gain (K_m) was derived from the system's DC gain and the time constant (T_m) was found by identifying the -3 dB corner frequency on the magnitude plot.

Baseline P, PD, and PID controller gains were calculated analytically using the experimentally determined system parameters. These gains served as the starting point for tuning within a non-linear Simulink model. When modeling the controller gains in Simulink, the linear model controllers show much higher transient aggression, visualized as very visible overshoot due to the absence of real-world nonlinear effects such as saturation limits and dead zone. In contrast, the nonlinear model is able to account for these real-world effects, allowing the Simulink simulation to more accurately reflect the measured dynamics of the system observed in physical testing. This refinement ensures that the simulated responses during further tuning closely match the real-world behavior within the operating constraints ($M_p \leq 6\%$, $t_s \leq 5s$).

To accurately align the simulated and measured system behaviors with the non-linear Simulink model, the real-world elements of the model were incorporated directly using block components. A Saturation block was applied to the output of the sum block of the PID_K blocks feeding into the Motor Transfer Function. This block replicates the physical saturation limits of the motor, which was determined experimentally to be -4.81 V for the lower bound and $+5.81$ V for the upper bound. These bounds are under the no added mass setup. This clips any control effort beyond those ranges in the simulation, reflecting the real operation. A dead zone block was also implemented in a similar manner, to model the experimentally measured region where small inputs do not yield any physical response due to actuator friction and actuator threshold effects. The bounds for this data were collected in a similar manner to the saturation data, with the dead zone bounds being between -0.078 V and $+0.0488$ V. The controller gains were then manually tuned within this high-fidelity model with the objective of meeting the required performance constraints.

To manually tune the system, it is important to follow certain heuristics. Generally, increasing the proportional gain increases overshoot and decreases settling time. Increasing the derivative gain increases settling time but decreases overshoot. And finally, increasing integral gain decreases the steady state response error. The tuning process can very reliably achieve a system within specifications when these steps are followed.

Results

The experimental investigation yielded two key outcomes: a precise, validated model of the DC motor and a tuned PID controller that meets the required performance requirements under specific operating conditions. The initial system identification successfully quantified the motor's linear operating range, the impact of added mass, and a reliable set of parameters for the system model. This accurate model was foundational to the development of a controller whose simulated performance in a non-linear model closely matched the physical system's response.

System Identification and Parameterization

A critical first step in modeling the system was to experimentally define its linear operating range, as the validity of the first-order transfer function is confined to this region. The non-linear boundaries of the motor's response were determined as shown in *Figure 1*. Analysis of the local gain (left panel) identified that the system enters saturation beyond approximately -4.8 V and 5.81 V. A magnified view of the response near the origin (right panel) revealed a dead zone between -0.078 V and 0.049 V, a phenomenon attributed to static friction.

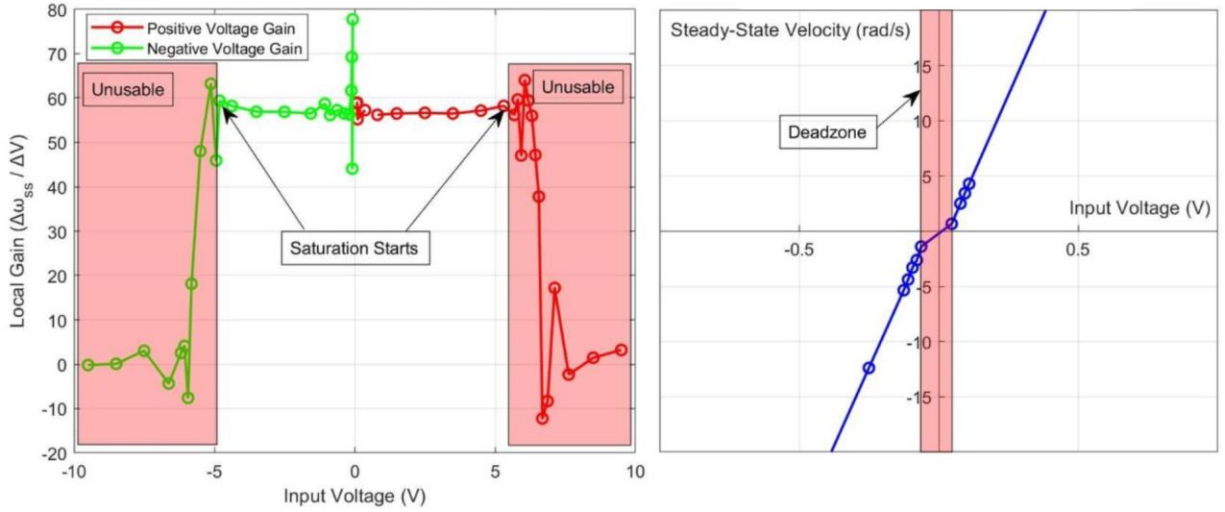


Figure 1: Determination of the motor's non-linear boundaries. Analysis of the local gain identifies the onset of saturation (left), while a magnified view defines the dead zone (right).

With the linear operating range established, a York regression was fit to the data within this valid region, as shown in Figure 2. This robust, time-domain analysis yielded a motor gain constant of $K_m = 55.9967 \pm 0.0612$ rad/s/V ($K=2$). As illustrated by the $K=2$ prediction band, all experimental data points used for the fit fall within the 95.4% confidence interval, confirming the high quality of the linear model.

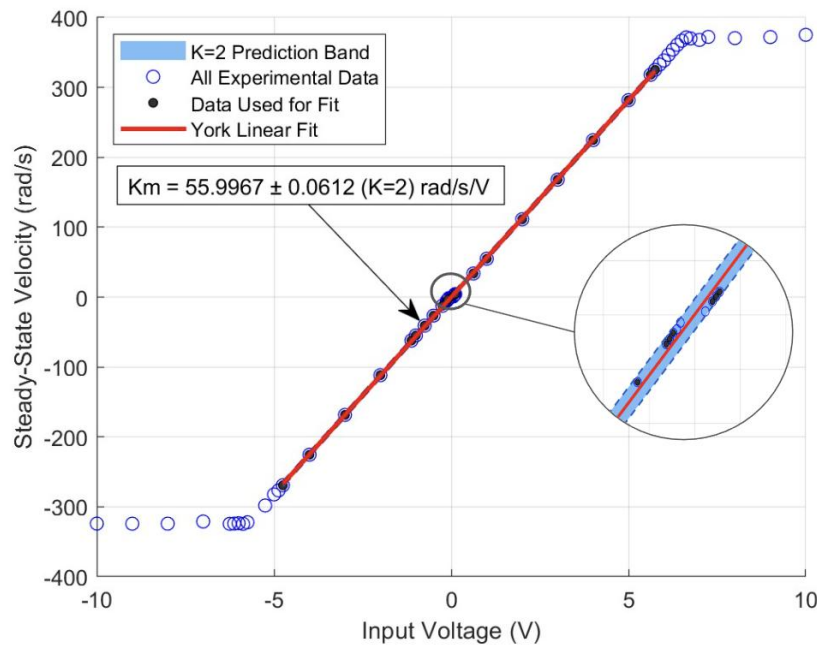


Figure 2: York linear regression fit to the time-domain data within the experimentally determined linear operating range, including the $K=2$ prediction band.

The system was also characterized in the frequency domain to validate the parameters, with the results summarized in the Bode plot in Figure 3. This analysis produced a highly consistent motor gain of $K_m = 57.1021 \pm 0.0721$ rad/s/V ($K=2$) rad/s/V and provided two estimates for the

time constant: $T_m = 0.07683 \pm 0.00528$ (K=2) s from the -3dB magnitude cutoff and $T_m = 0.09434 \pm 0.00018$ (K=2) s from the -45° phase shift.

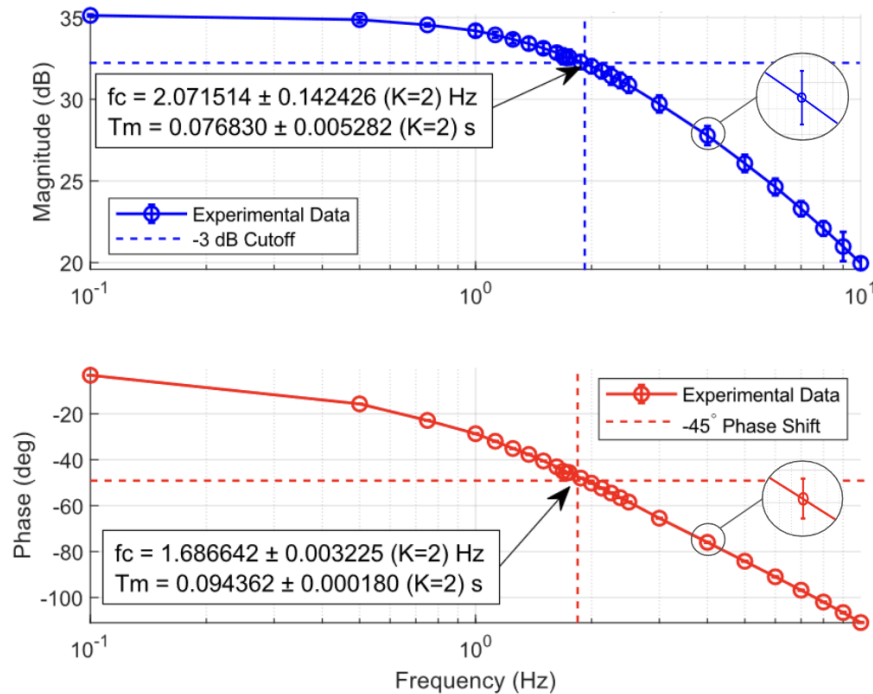


Figure 3: Bode plot of the motor's frequency response. The analysis provides values for the motor gain (K_m) and two estimates for the time constant (T_m) based on magnitude and phase data.

For the final model, the time-domain value of $K_m = 55.9967 \pm 0.0612$ rad/s/V was selected for its robustness, as it is derived from a regression across the motor's entire useful operating range. Conversely, the time constant, $T_m = 0.07683 \pm 0.00528$ s was selected from the frequency-domain analysis, as this method provides a more comprehensive characterization of the motor's dynamics than relying on individual time-domain tests. The final value was taken specifically from the magnitude plot, as its -3dB corner frequency is a more robust feature than the phase plot's -45° crossing, which is susceptible to unmodeled system delays.

Effect of Additional Flywheel Mass

The effect of adding a 104 g mass was investigated to understand how increased inertia impacts system performance.

Table 1: Effect of added mass on time-domain motor constants

Parameter	No Added Weight	Added Mass (+104g)
Motor Gain, K_m	55.9967 rad/s/V	56.21 rad/s/V
Time Constant, T_m	0.0858 s	0.2162 s

The results, summarized in *Table 1* above, align perfectly with theoretical expectations. The motor gain (K_m), primarily a function of electrical properties, remained virtually unchanged. In contrast, the time constant (T_m) more than doubled, reflecting the physical reality that a higher-inertia system takes longer to accelerate and respond to inputs.

Controller Tuning Process and Effects of Gains

The initial, analytically calculated gains provided a theoretical baseline, but manual tuning was essential to optimize performance and meet the required specifications. This tuning process was conducted within the high-fidelity non-linear Simulink model. The performance of the initial, analytically calculated gains was first evaluated in the non-linear simulation, with the results summarized in *Table 2*. While the PD and PID controllers successfully met the performance specifications, their settling times were very long (over 4 seconds), thus failing to leverage the full dynamic capability of the motor. The baseline P controller failed to meet the steady-state error requirement. This analysis confirmed that while the analytical calculations provide a stable starting point, manual tuning is essential to achieve an aggressive, high-performance response.

Table 2: Performance of initial, analytically calculated gains in the non-linear Simulink model. Values in green denote being within specification. Values were measured at a setpoint of 4 radians.

Controller	Model	Max Overshoot [%]	Settling Time [s]	Steady-State Error [rad]	Meets All Specs?
P	Nonlinear	0.64722	4.4439	0.2619	x
PD	Nonlinear	0.00	4.4315	0.00533	✓
PID	Nonlinear	0.018725	4.423	0.00079	✓

The dramatic improvement from this tuning process is visualized in *Figure 4*. The final tuned gains produce a response that is significantly faster and more controlled than the initial calculated baseline, demonstrating the value of manual optimization techniques.

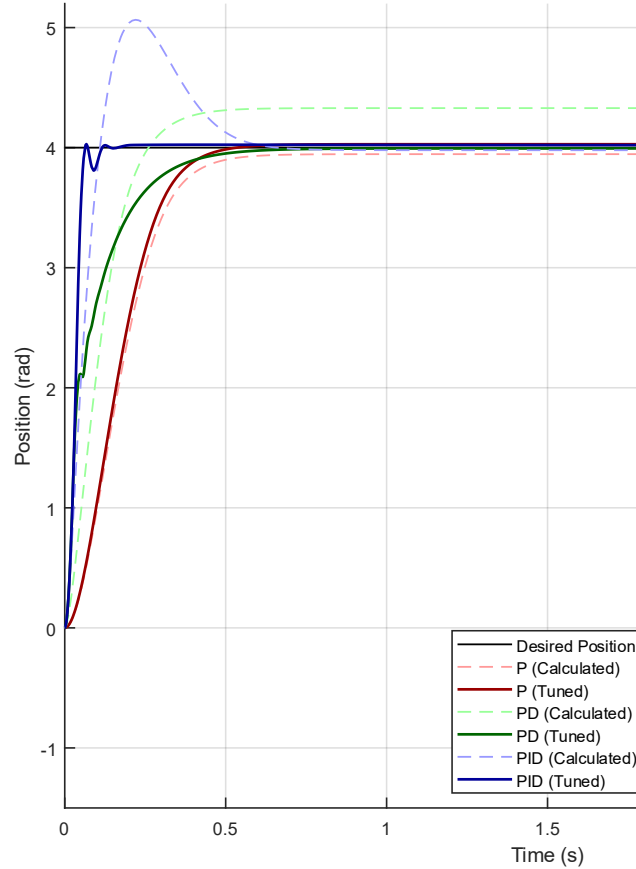


Figure 4: Comparison of controller performance in the non-linear model using the initial analytically calculated gains versus the final manually tuned gains.

The final tuned gain values used for all subsequent analysis are listed in *Table 3*. The tuning process involved iteratively adjusting the gains to achieve the desired response requirement: the proportional gain (K_p) was increased to speed up the system, the derivative gain (K_d) was increased to dampen overshoot, and the integral gain (K_i) was increased to eliminate steady-state error.

Table 3: Final manually tuned gain values

Controller	K_p	K_i	K_d
P	0.13	0	0
PD	7.5	0	0.8
PID	7.5	0.15	0.8

Model Accuracy and Validation

The analysis demonstrates that the non-linear model is significantly more accurate and essential for this application. With the aggressive gains needed to meet the performance specifications, the linear model, failing to account for physical saturation limits, predicted a wild oscillatory response. In contrast, the non-linear simulation and the real-world data show a stable, controlled response, as seen in *Figures 5 and 6*. The fitting confirms that the non-linear model is an accurate and reliable tool for predicting the system's true behavior. The minor discrepancies between the non-linear simulation and the real world are likely due to small, unmodeled effects such as viscous friction, and they are well within acceptable limits.

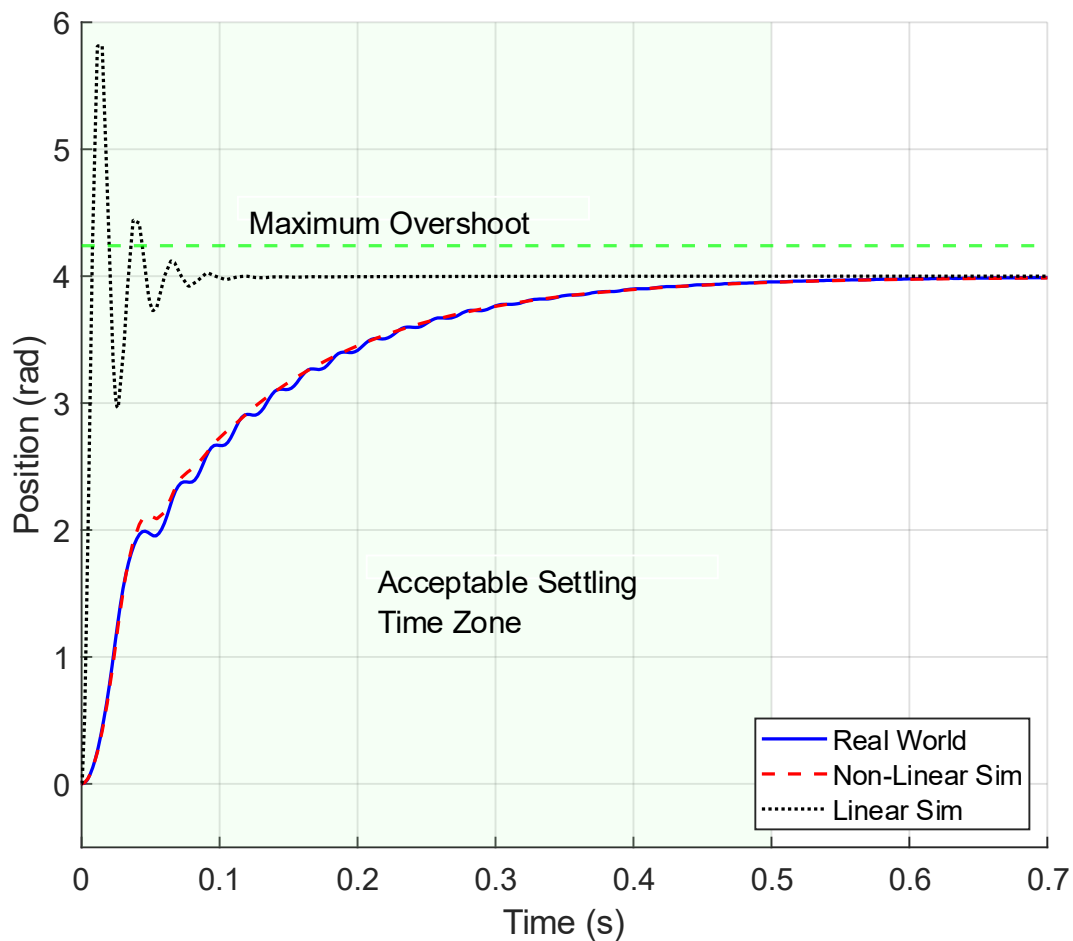


Figure 5: Performance of the tuned PD controller, comparing the real-world response to the non-linear and linear simulation models.

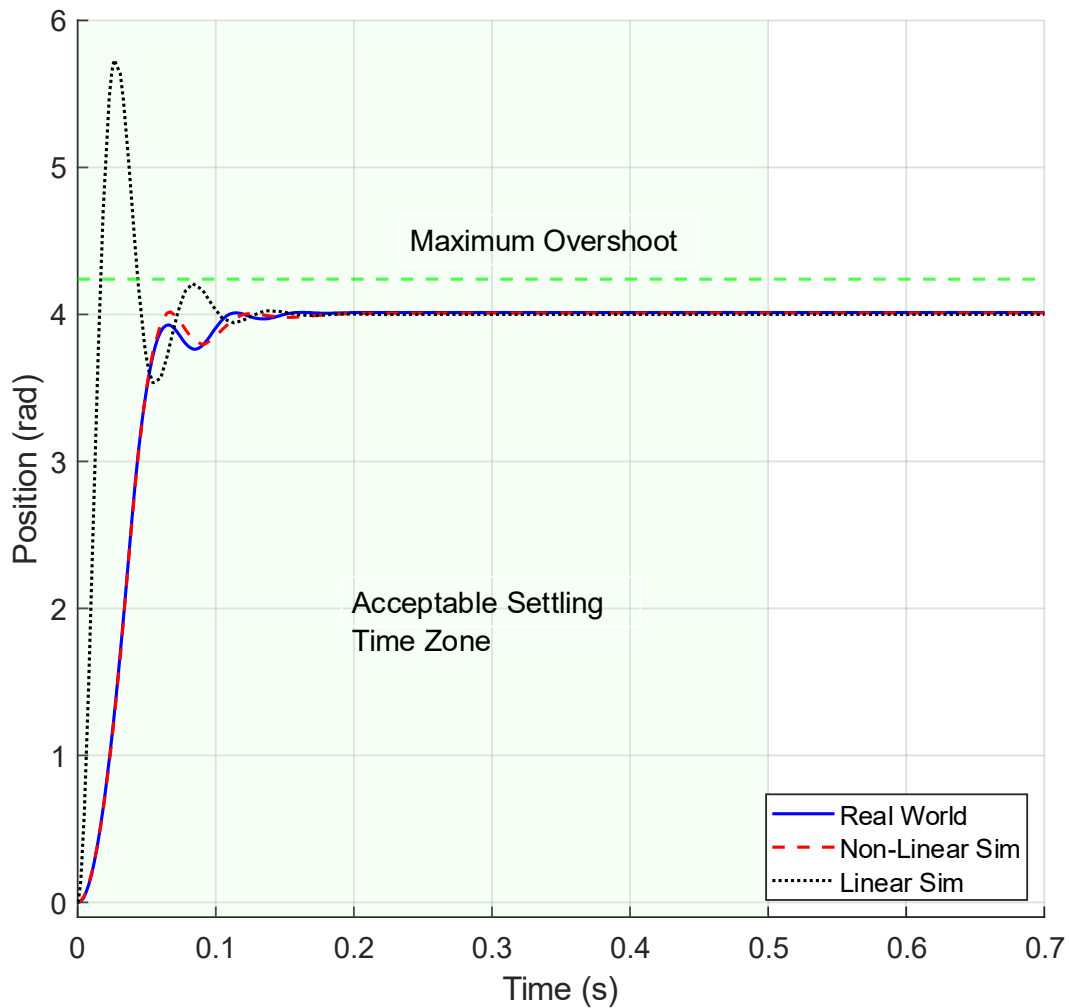


Figure 6: Performance of the tuned PID controller for a 4 rad command, comparing the real-world response to the non-linear and linear simulation models.

Performance Limitation

The final analysis identified a performance limitation at small command amplitudes. Figure 7 shows that for target positions below approximately 1.0 radians, the physical motor's response becomes inconsistent and begins to deviate from the simulated model, exhibiting higher overshoot. This behavior is likely due to the increased influence of stiction and encoder quantization on small movements. This finding establishes a reliable operating boundary for the controller; while the controller meets all specifications, its predictable, high-performance operation is confined to target positions of 1.0 radians or greater.

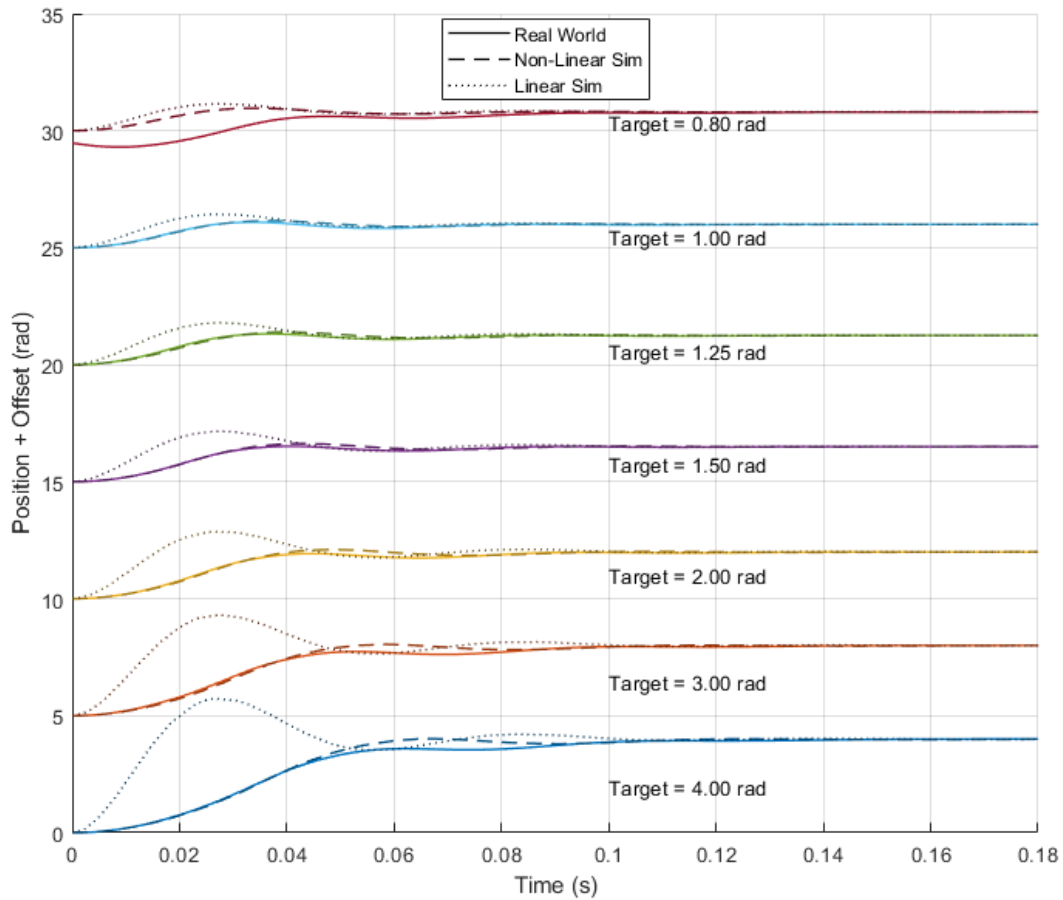


Figure 7: Comparison of simulated vs. real-world PID performance across a range of target positions, showing increased deviation and overshoot for commands below 1.0 rad. Setpoint for each trial is 4 radians.

Further Improvements

The motor can be used as is with PID as the default strategy and PD as a low-overshoot alternative, provided operation remains inside the linear window and commanded steps are above the 1.0 rad threshold or otherwise compensated. Practical safeguards should be incorporated in firmware: dead-zone compensation (bias or preload) to ensure repeatable initiation of motion, and output limiting to avoid drive saturation. Because load inertia directly affects T_m , the derivative and proportional gains should be adjusted when the payload changes. For example, if overshoot becomes too high, increasing K_d (and if needed decreasing K_p) is often sufficient to lower overshoot to be within 6%.

Further sensor and equipment upgrades could also improve the efficiency of the tuning process. On the identification side, cleaner low-frequency sinusoid excitation and filtering of encoder velocity would sharpen the corner-frequency estimate and reduce uncertainty in T_m . On the control side, adding a setpoint scheduler that scales gains for small moves will improve repeatability below 1.0 rad; a two-stage controller (PD for approach, PID for final trim) can also enhance precision without sacrificing speed. If future revisions permit hardware changes, a higher-resolution encoder would reduce quantization, further increasing confidence that the specified transient and steady-state metrics will be met across operating conditions.

Conclusion

Experimental identification produced a reliable dynamic model of Webb Industries' DC motor for position control. Across time and frequency-domain testing, the motor gain was consistent at $K_m \approx 55.967 \pm 0.061 \text{ rad}\cdot\text{s}^{-1}\cdot\text{V}^{-1}$ (K=2), and the nominal time constant was $T_m \approx 0.082 \pm 0.003 \text{ s}$ (K=2), increasing to $T_m \approx 0.216 \pm 0.012 \text{ s}$ (K=2) with a +104g load. Dead-zone and saturation mapping established bounds at -0.078 to $+0.0488 \text{ V}$ (dead zone) and -4.81 V and $+5.81 \text{ V}$ (saturation); operation outside this range produced unreliable local gain and should be avoided. In addition, responses to small setpoint changes below 1.0 rad were inconsistent and are not recommended for qualification testing or fine positioning without compensation. Within the identified window, controllers tuned from the measured parameters satisfied Webb Industries' criteria of $M_p \leq 6\%$, $t_s \leq 0.5 \text{ s}$, and $|e_{ss}| \leq 0.25 \text{ rad}$. In nonlinear simulations reflecting measured limits, PID yielded the fastest compliant settling with negligible steady-state error, PD achieved a similarly accurate response with lower overshoot and slightly longer settling time.